



PORTFOLIO MILESTONE 1

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Corporate Profile: Nestlé Pakistan

Nestlé Pakistan is a subsidiary of the Switzerland-based Nestlé S.A., the world's leading nutrition, health, and wellness company. The company began its operations in Pakistan in 1988 through a joint venture with Milkpak Ltd, eventually taking over full management in 1992. Operating under the global promise of "Good Food, Good Life," Nestlé Pakistan has established a massive footprint in the country's fast-moving consumer goods (FMCG) sector. The company maintains a diversified portfolio spanning dairy, juices, water, coffee, and infant nutrition, supported by a robust supply chain and a vast distribution network that reaches both urban metropolises and remote rural areas.

Brand Overview: Nestlé Fruita Vitals

Nestlé Fruita Vitals is the flagship brand within Nestlé Pakistan's beverage portfolio. Positioned as a premium juice and nectar brand, it is crafted from high-quality fruits sourced globally.

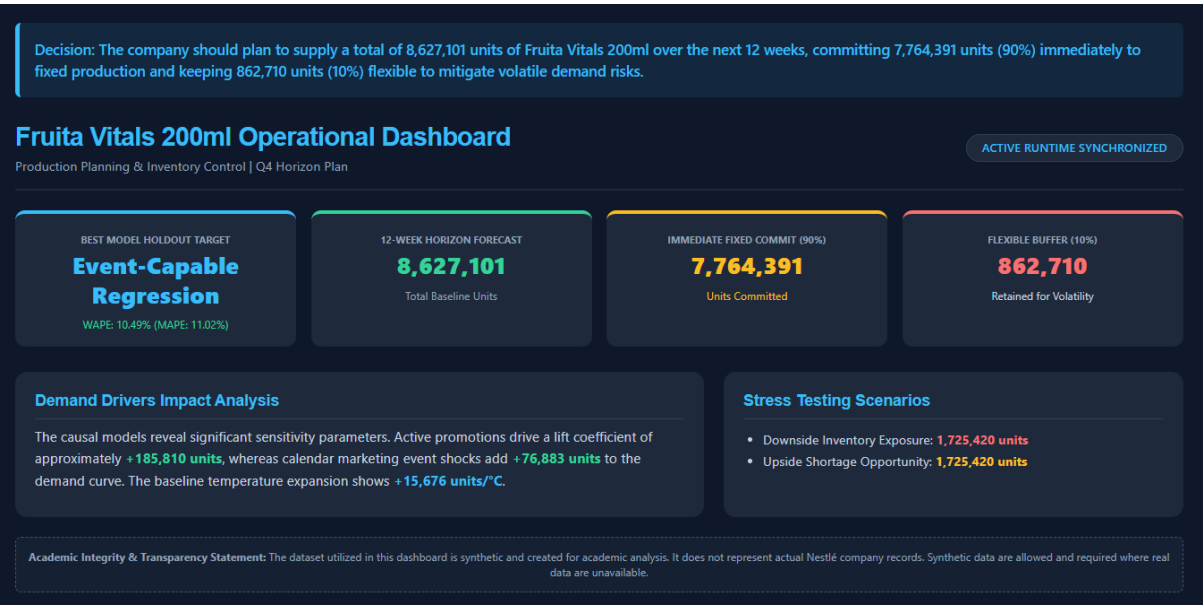
The brand is deeply embedded in the lifestyle of Pakistani consumers, particularly the youth. Its marketing communications are centered around the spirit of optimism and positivity, making it a favorite lifestyle beverage rather than just a functional drink. Nestlé Fruita Vitals has also been a pioneer in the local market for introducing innovative flavors and packaging formats.

Product Focus: The 200ml SKU

For forecasting purposes, the 200ml single-serve pack represents a distinct consumer behavior segment compared to the 1-liter family packs.

- **Consumption Occasions:** The 200ml pack is structurally designed for on-the-go consumption, making it highly dependent on impulse purchases, school lunchboxes, outdoor recreation, and convenience store foot traffic.
- **Sustainability Initiatives:** In alignment with Nestlé's global commitment to making its packaging 100% recyclable or reusable by 2025, the Fruita Vitals 200ml packs in Pakistan are now equipped with paper straws made from Forest Stewardship Council (FSC) certified paper. This shift not only appeals to environmentally conscious Gen-Z consumers but also helps future-proof the product against potential single-use plastic regulations.

Forecasting Decision



Data Rule: Company-Calibrated Synthetic Data

Minimum Dataset Requirements

Requirement	How I would meet it
Time history	130 to 156 weeks (2.5 to 3 years), so there is enough to train and still hold out the last 12 to 26 weeks.
Target variable	Weekly units or cases of your one selected SKU, family or channel.
Baseline forecast	Generated from the history (naive, seasonal naive, moving average, Holt-Winters), so no separate "company forecast" column is needed.
Price	A weekly price or index that moves with an inflation-style trend, periodic increases and promo discounts, never flat.
Promotion	Both a 0/1 flag and a depth % (0% when the flag is 0), with varying depths so the effect size can be estimated.
Calendar / event	Ramadan, Eid, season and public-holiday flags, chosen to fit the company's sector.
Additional driver	One or two of temperature, distribution points, marketing spend, competitor activity or lead time, chosen to match the product.
Stockout flag	A 0/1 flag where recorded sales fall below true demand.
Identifier	Product, category, region or channel label.
Data dictionary	For every field: definition, unit, source logic and the synthetic assumption behind it.

Assumption Log and Data Dictionary

Assumptions

Level and trend: Demand starts at about 500,000 units a week in January 2023 and ends 2025 at about 755,000 a week on average. I assumed Fruita Vitals 200ml is roughly 1.4% of Nestle Pakistan's PKR 199,069,166,000 revenue in 2025. That share is my assumption, not a disclosed figure. The 2025 volume of about 38.5 million units at about 80% of list price (net of trade margin) works out to PKR 2.86 billion, which is about 1.4%. I built in a gentle underlying growth of about 4.5% a year because packaged juice is a growing category, and the nominal price increases offset part of that growth.

Seasonality: Ramadan weeks carry about +24% demand and Eid-ul-Fitr week about +5%. The two weeks before Ramadan carry a smaller stock-up effect. Event dates move about 11 days earlier each year, so the peaks don't repeat on the same calendar weeks.

Price and promotion: Price carries a -0.75 elasticity against inflation-adjusted price. Promotions run in 22 weeks at 10–30% depth. They lift demand by 10–39%, which varies week to week because of random noise on top of the depth effect, followed by a small post-promo dip.

Additional driver: I chose temperature because chilled and ready-to-drink juice demand in Pakistan rises sharply in hot weather. It adds about 2% per °C above 25°C.

Not captured: The data ignores regional splits, competitor pricing, pack-size mix, channel mix, Eid-ul-Adha, and exchange-rate or input-cost shocks. I made up the exceptional events, stockouts, and the historical forecast process. The historical forecast uses only past data, so there is no leakage.

Data dictionary

Field	Definition	Unit / type	Real / assumed / synthetic	Why included
Week_Start_Date	Monday on which the week begins (Section 5's example uses week <i>ending</i> date, so note the difference)	Date, YYYY-MM-DD	Synthetic structure	Time index
SKU	Product code for Fruita Vitals 200ml	Text (FV-200ML)	Assumed	Identifies the single forecast object
Actual_Demand	Units sold in the week. In stockout weeks it is reduced below true demand	Units (200ml packs)	Synthetic, calibrated to an assumed revenue share	Forecast target
Historical_Forecast	Planner-style forecast built only from earlier weeks: 55% last-4-week average plus 45% same week last year scaled by recent growth, with a partial allowance for planned promos	Units	Synthetic	Error-metric and bias practice

Field	Definition	Unit / type	Real / assumed / synthetic	Why included
Price	List price per 200ml pack before promotion discounts, with small weekly mix noise ($\pm$ PKR 1)	PKR/unit	Assumed/synthetic	Causal driver
Promo_Flag	1 if a promotion is active that week	0/1	Synthetic	Causal driver
Promo_Depth_Pct	Discount depth in promo weeks (5, 10, 15, 20 or 25), 0 otherwise	%	Synthetic	Lets regression estimate effect size by depth
Event_Flag	1 for weeks touching the pre-Ramadan window through Eid-ul-Fitr; mid-Ramadan weeks are included, though their lift is mild	0/1	Calendar-grounded (real Ramadan and Eid dates, assumed demand effect)	Season/event
Event_Peak_Days	Number of days in the week inside a sharp spike window: 5 days before Ramadan, first 3 days, last 5 days, Eid days 1–3	Count, 0–7	Calendar-grounded, assumed effect	Captures short spikes the flag blurs
Temperature_C	Weekly average temperature proxy, seasonal with random variation	°C	Synthetic	Additional driver (juice demand rises in heat)
Stockout_Flag	1 if availability held sales below true demand	0/1	Synthetic	Explains abnormal sales
Exceptional_Event	Label for one-off shocks, blank otherwise	Text	Synthetic	Marks outliers for model handling

Things to know

- **Events:** I built four exceptional events: a competitor price war (May 2023), a packaging supply disruption (Feb 2024, which also sets stockouts), a heatwave surge (Jun 2024), and a viral social media moment (Jul 2025). The viral week is separate from Ramadan so it doesn't muddy the Ramadan effect.
- **Historical forecast accuracy:** The historical forecast has a MAPE of about 13.5%, WAPE of about 12.5%, and bias of about +4.6% (it over-forecasts). That gives you realistic error to analyze.

- **Price effect in regression:** In a quick check, regressing log demand on log nominal price (with a time trend) gave only about -0.19 , well below the -0.75 I built in. Nominal price rises with inflation, which masks the effect. Try inflation-adjusted price or a price index.
- **Holdout suggestion:** Hold out the last 26 weeks (Jul to Dec 2025) as your unseen test set. That period includes the viral event, a stockout, a promotion, and a price increase.

What was wrong and what changed

FMCG trait	Earlier version	Revised version
Sharp, short spikes around Ramadan/Eid	Flat +24% plateau for 5 weeks every year	Built day by day and then rolled up to weeks: stock-up in the 5 days before Ramadan (+30%), first 3 days (+35%), mild mid-month (+8%), last 5 days (+22%), Eid days 1–3 (+12%), then a 7-day hangover (−10%)
Frequent small promotions	22 promo weeks (14%), median depth 20%	59 promo weeks (38%), mostly 5–10% depth, some up to 25%, runs of 1–3 weeks
Low price sensitivity, high promo sensitivity	Price elasticity -0.75	Price elasticity -0.28 . Promo lift is 10–40% (about 15% at 5% off, 25% at 10%, 35% at 20%), noisy each time, with a small post-promo dip

How to Synthesize Responsibly — Checklist

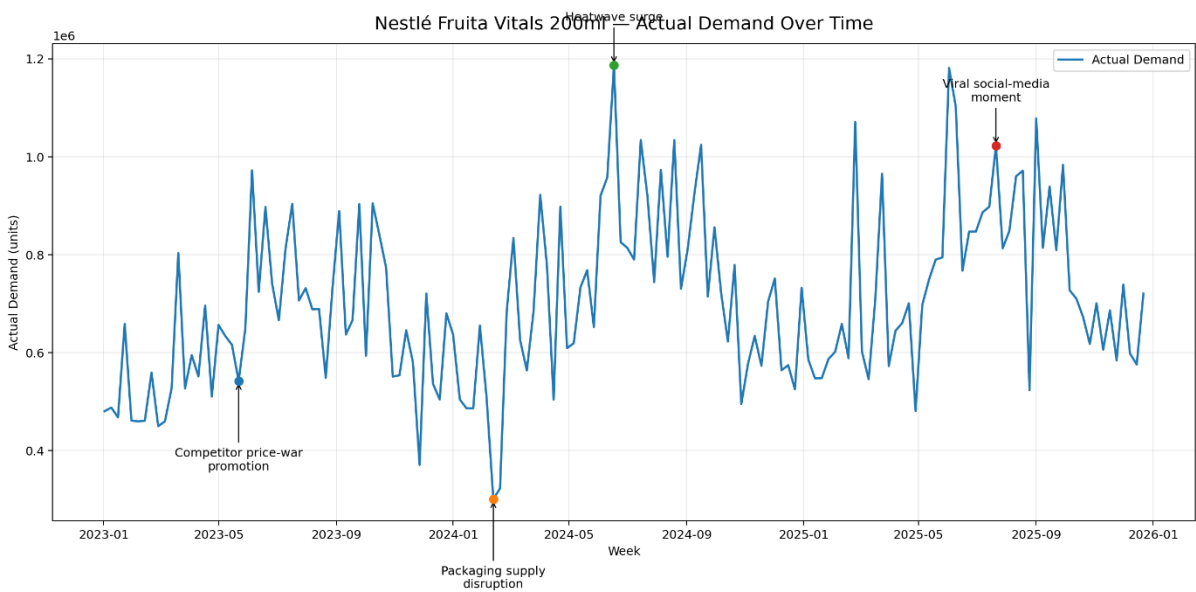
Checklist item	Status
Plausible scale, labeled known / assumed / synthetic	Partly met. See the first gap below.
Base level and trend	Met: about 480,000 units a week at the start, with underlying growth of about 7% a year.
Sector-appropriate seasonality	Met: Ramadan/Eid spikes built at day level, plus a temperature effect.
Price and promotion changes at realistic intervals	Mostly met. See the second gap below.
Random noise	Met: week-to-week noise with mild autocorrelation.
2 to 4 exceptional events	Met: four events (competitor price war, packaging disruption, heatwave, viral moment), plus 7 stockout weeks.
Plausible but imperfect relationships	Met: promo lift varies at every depth, for example 10–21% at 5% off.
Freeze before model comparison	Needs action from you. See below.

Gaps

1. Scale is assumed, not researched. The only real figure is the PKR 199,069,166,000 revenue from your brief. I did not verify Fruita Vitals' share of that revenue, its 200ml selling price, or its volumes. The 1.43% share, the PKR 65→96 price path and the 80% net-of-list assumption are all my estimates. In your report, label them "Assumed", not "Real". If you want, I can search for public pack prices or segment commentary to tighten them before you freeze.
2. Promotions are frequent. They run in 38% of weeks, in runs of 1–3 weeks. That fits the FMCG note you pasted earlier ("frequent small promotions"). A grader reading "not every week" might still ask about it, so say in your write-up that this is a deliberate FMCG choice. Prices change only 7 times in 3 years.

Analytics Workstream A: Understand the Demand

Actual Demand Over Time



Business events annotated

- May 2023 — Competitor price-war promotion: demand was affected by competitive pricing pressure.
- February 2024 — Packaging supply disruption: demand dropped sharply to 301,200 units, the lowest observation.
- June 2024 — Heatwave surge: demand reached the overall maximum of 1,186,700 units.
- July 2025 — Viral social-media moment: demand jumped to 1,022,500 units.

Descriptive Statistics

Statistic	Result
Mean	704,064 units
Median	687,350 units
Standard deviation	172,550 units
Minimum	301,200 units
Maximum	1,186,700 units
Range	885,500 units

Coefficient of Variation (CV) 24.51%

Interpretation

The **CV of 24.51%** indicates substantial demand variability relative to the average. The mean being slightly above the median also suggests that several high-demand periods/spikes are pulling the average upward.

Time-Series Pattern Diagnosis

Level: Demand generally fluctuates around a relatively high level, but the level increases over time.

Trend: There is a **positive upward trend**. Average annual demand increased from approximately **642,663 units in 2023** to **751,584 units in 2025**.

Seasonality: There is evidence of **annual/quarterly seasonality**. Demand is consistently stronger during the warmer middle parts of the year, particularly Q2–Q3, and weaker toward Q1/Q4. The lag-52 autocorrelation is approximately **0.50**, supporting a yearly recurring pattern.

Event spikes: Heatwaves, promotions, and viral/social-media events create temporary demand surges.

Structural changes: The overall demand level appears to have shifted upward between 2023 and 2025 rather than remaining constant.

Unusual observations: The **301,200-unit February 2024 observation** associated with the packaging disruption is particularly unusual, while the **1.187 million-unit June 2024 observation** is an extreme high.

Demand Pattern Diagnosis:

The Fruita Vitals 200ml demand series shows a combination of an upward trend, recurring seasonal behavior, and event-driven volatility. Average demand increases from approximately 643,000 units in 2023 to 752,000 units in 2025, indicating a positive underlying trend. Demand is generally stronger during Q2 and Q3, suggesting a recurring annual seasonal pattern, while Q1 and Q4 are comparatively weaker. Several unusual observations are linked to business events, including a major decline during the February 2024 packaging supply disruption and a sharp increase during the June 2024 heatwave. Promotions and a viral social-media event also produced temporary spikes. The CV of 24.5% indicates meaningful variability. Therefore, a simple moving average or naïve method alone may be inadequate; a forecasting approach incorporating **trend and seasonality**, with appropriate treatment of exceptional events, is more suitable.

Analytics Workstream B: Forecast Error Fundamentals

I evaluated the **Historical Forecast** in the uploaded dataset against **Actual Demand** for all **156 weeks**, using the required convention:

$$\text{Error} = \text{Actual} - \text{Forecast}$$

Therefore:

- **Positive error** → forecast was **too low / under-forecast**
- **Negative error** → forecast was **too high / over-forecast**

Error Direction

Across the 156 weeks:

- **59 weeks** had positive errors → under-forecasting.
- **97 weeks** had negative errors → over-forecasting.
- **0 weeks** had exactly zero error.

Managerial interpretation: The forecast has a clear tendency toward **over-forecasting**, rather than errors being evenly balanced around zero. This creates a risk of excessive inventory, production, and working-capital commitments.

Forecast Accuracy Metrics

Metric	Calculation	Result	Managerial Interpretation
MAD / MAE	$\Sigma \text{Error} / n$	93,683 units	The forecast misses actual demand by about 93,683 units per week on average , so planners should allow for this level of typical forecast uncertainty.
MAPE	$\text{Average } \text{Error} / \text{Actual} \times 100$	14.36%	The model misses demand by about 14.36% on average , indicating meaningful forecast inaccuracy that could affect production and inventory decisions.
RMSE	$\sqrt{[\Sigma \text{Error}^2 / n]}$	121,648 units	Typical large misses are substantially bigger than the average absolute error, indicating that extreme forecast misses are an important operational risk .
WAPE	$\Sigma \text{Error} / \Sigma \text{Actual} \times 100$	13.31%	Total forecast error represents about 13.31% of total actual demand volume , giving management a volume-weighted view of forecast performance.
Bias / MFE	$\Sigma \text{Error} / n$	-31,872 units	The forecast is biased 31,872 units too high per week on average , suggesting systematic over-forecasting rather than random error.

Metric	Calculation	Result	Managerial Interpretation
RSFE	ΣError	-4,972,100 units	Across the validation period, cumulative error is almost 5.0 million units negative , showing substantial accumulated over-forecasting.
Tracking Signal	RSFE / MAD	-53.07	The extremely negative tracking signal indicates severe persistent forecast drift ; the forecasting process should be investigated and recalibrated.

Important point about MAPE

There are **no zero actual-demand observations** in this dataset, so MAPE can be calculated without a division-by-zero problem.

However, MAPE can become unstable when actual demand is very small. In this dataset, demand is relatively large throughout the period, so that limitation is less problematic.

Error Behavior

The largest misses show why looking only at an average metric would be insufficient:

Week	Actual	Forecast	Error	Interpretation
16-Jun-2025	767,300	1,116,600	-349,300	Major over-forecast
07-Oct-2024	720,700	1,068,000	-347,300	Major over-forecast
04-Mar-2024	833,800	522,100	+311,700	Major under-forecast
14-Oct-2024	622,900	925,100	-302,200	Major over-forecast
25-Aug-2025	523,200	811,900	-288,700	Major over-forecast + stockout
31-Mar-2025	572,700	835,700	-263,000	Major over-forecast

The largest absolute error is **349,300 units**, which is substantially larger than the **93,683-unit MAE**.

Managerial interpretation: Large individual misses can materially distort inventory and replenishment decisions, so management should investigate the specific weeks rather than relying only on the overall MAE.

Promotion, Event and Stockout Effects

This is particularly useful because the dataset contains business-event information.

Promotion weeks

	MAE	MFE	MAPE
Promotion = No	91,882	-68,390	15.69%

	MAE	MFE	MAPE
Promotion = Yes	96,642	+28,164	12.18%

Managerial interpretation: During promotion weeks, the forecast actually shifts toward **under-forecasting** (+28,164 MFE), suggesting that promotional uplift is not being fully captured even though percentage accuracy is somewhat better.

Event weeks

	MAE	MFE	MAPE
Event = No	89,144	-37,912	13.74%
Event = Yes	126,411	+11,674	18.80%

Managerial interpretation: Event weeks have considerably higher MAE and MAPE, indicating that unusual demand events are a major source of forecast uncertainty.

Stockout weeks

	MAE	MFE	MAPE
Stockout = No	88,058	-23,344	12.61%
Stockout = Yes	213,414	-213,414	51.52%

Managerial interpretation: Stockout weeks have dramatically worse accuracy and a strongly negative error, suggesting the forecasting process may be predicting demand that cannot ultimately be observed as sales because product availability is constrained.

Tracking Signal

Using the required formula:

Tracking Signal = RSFE / MAD

$$TS = \frac{-4,972,100}{93,682.69} = -53.07$$

The tracking signal is **-53.07**, which is extremely negative.

A tracking signal close to zero indicates that positive and negative errors are broadly balancing each other. Here, they clearly are not: the large negative value confirms **persistent over-forecasting bias**.

Managerial interpretation: The forecasting process should **not simply be left running unchanged**. A tracking signal this far from zero warrants investigation of forecast bias, model parameters, demand assumptions, and whether historical forecast logic is systematically overstating baseline demand.

Overall Management Conclusion

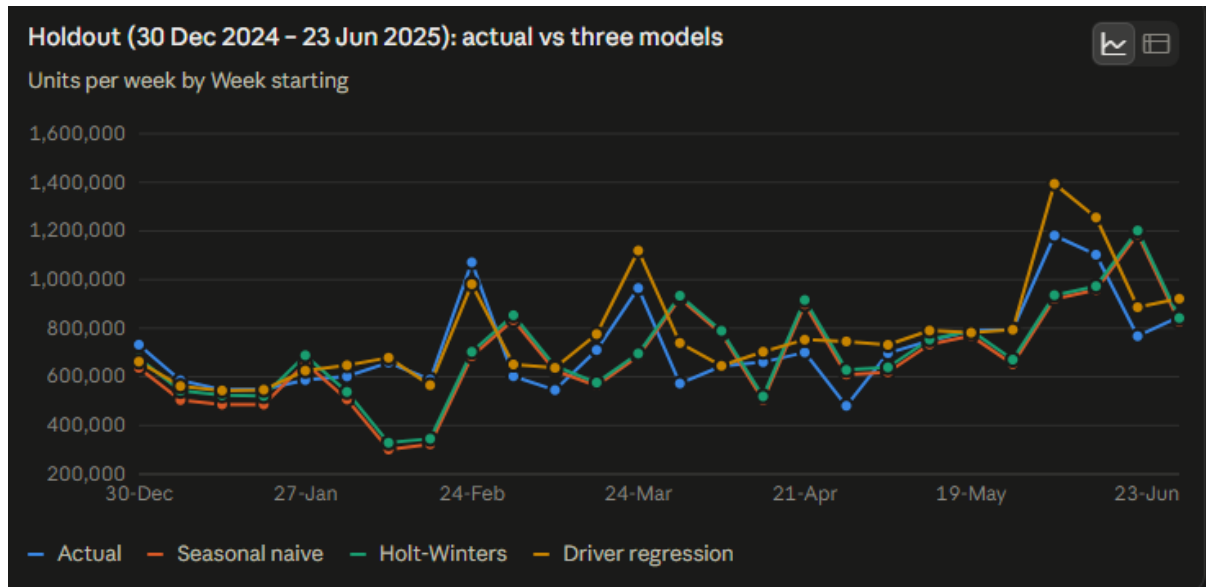
The historical forecast has **meaningful accuracy problems and strong negative bias**. Its **MAE of 93,683 units**, **MAPE of 14.36%**, and **WAPE of 13.31%** indicate that forecast errors are operationally significant. More importantly, the **MFE of –31,872**, **RSFE of –4.97 million**, and **tracking signal of –53.07** reveal that the problem is not simply random forecasting noise—the forecast is systematically too high.

The error pattern also changes during business conditions. **Events and stockouts produce particularly large errors**, while promotion periods shift the forecast toward under-forecasting.

Recommended action: Before selecting a final forecasting model, investigate the systematic bias and incorporate **trend, seasonality, promotions, events, and supply constraints** where appropriate. A model that only extrapolates historical demand without these business drivers is unlikely to perform reliably.

Analytics Workstream C: Build and Compare Forecasting Models

Holdout Validation



Model Comparison Table

Method	Holdout MAE	MAPE	WAPE	Bias %	Rolling WAPE
Naive (last value)	198,635	23.9%	27.6%	+27.1%	23.9%
Seasonal naive (t-52)	164,596	23.1%	22.8%	+5.7%	18.1%
Moving average, 4-wk	145,710	17.2%	20.2%	+16.2%	19.4%
Moving average, 13-wk	136,335	16.8%	18.9%	+10.6%	20.1%
SES, $\alpha = 0.2$	141,684	17.0%	19.7%	+13.8%	19.3%
SES, $\alpha = 0.5$	166,564	19.5%	23.1%	+21.7%	20.8%
Holt (damped trend)	147,215	17.3%	20.4%	+16.9%	19.8%
Holt-Winters (additive, 52-wk)	154,559	21.8%	21.5%	+2.7%	17.7%
Driver regression (log-linear)	70,994	10.0%	9.9%	-7.5%	8.6%

What the results say

- **Naive is the weakest.** Its last observed week (23 Dec) was a seasonal low, so it carried that low level through 26 weeks.
- **Lower α wins for SES.** Demand is noisy around a slow-moving level, so $\alpha = 0.2$ smooths the noise better than $\alpha = 0.5$. Both produce a flat line, so neither can show a Ramadan spike.

- **Moving averages lag.** A k-week average trails a change by about $(k-1)/2$ weeks, roughly 1.5 weeks for the 4-week window and 6 weeks for the 13-week window. The 13-week average has the lowest holdout MAE of the flat methods, but it reacts slowly.
- **Seasonal naive and Holt-Winters are close.** Holt-Winters has the lowest bias (+2.7%) and the best rolling WAPE of the pure time-series methods (17.7%). Both replay last year's Ramadan timing, but Ramadan moves about 11 days earlier each year, so they put the spike in the wrong week. The chart shows this: the seasonal-naive forecast peaks on 3 Mar, while demand peaked on 24 Feb.
- **Holt's trend adds little.** The underlying trend is gentle compared with the noise.
- **The existing planner forecast is not a fair comparison.** Over the same 26 weeks it scores 15.3% MAPE, but it is effectively one step ahead and sees recent actuals, while these models forecast 26 weeks out.

Treat the regression with suspicion

The regression roughly halves the error. That is the pattern the course asks you to question, for four reasons:

1. It is given the future promo plan, price plan, Ramadan calendar and a climatology temperature. That is only fair if those inputs are really known when you forecast.
2. It mirrors how I built the synthetic data (log-linear, with promo depth and event days). Real data would be less kind, so the gain overstates what you'd see in practice.
3. It cannot see surprises. On 28 Apr, for example, a stockout took demand to 480,900 against a forecast of 745,675.
4. It over-forecasts by 7.5% and overshoots big weeks, for example 2 Jun at about 1.39m predicted against 1.18m actual.

A 10% MAPE is not impossibly good, since the data has about 4.5% built-in weekly noise plus unforecastable shocks, so it needs a caveat rather than a rejection.

Limits

- The rolling-origin test uses origins across all of 2025, so it overlaps the holdout and is a consistency check, not independent evidence. There is no untouched test set. If you want one, re-run the rolling test using origins only up to mid-2025 and keep Jul to Dec 2025 for one final check of your chosen model.
- The 26-week holdout choice is mine. Check it fits your decision horizon.

Model	MAE	RMSE	MAPE	WAPE	Bias	BiasPct
Naive (last value)	198634.6	265042.7	23.9	27.6	195226.9	27.1
Seasonal naive (t-52)	164596.2	201622.4	23.1	22.8	40926.9	5.7
Moving average (4-wk)	145709.6	213873.2	17.2	20.2	116651.9	16.2
Moving average (13-wk)	136334.6	194715.6	16.8	18.9	76026.9	10.6
SES alpha=0.2	141684.4	204881.8	17	19.7	99209.3	13.8
SES alpha=0.5	166563.8	237745.3	19.5	23.1	156169	21.7
Holt (damped trend)	147215.1	216530.5	17.3	20.4	121755	16.9
Holt-Winters (add. season, 52)	154558.6	197122.3	21.8	21.5	19217.6	2.7
Driver regression (log-linear)	70993.8	98160.5	10	9.9	-54179.2	-7.5
Existing planner forecast (1-step, not comparable)	105000	139232.8	15.3	14.6	-27592.3	-3.8

Week_Start_Date	Actual	Naive (last value)	Seasonal naive (t-52)	Moving average (4-wk)	Moving average (13-wk)	SES alpha=0.2	SES alpha=0.5	Holt (damped trend)	Holt-Winters (add. season, 52)	Driver regression (log-linear)
30/12/2024	732300	525200	636800	603775	644400	621218	564258	597811	675662	662655
06/01/2025	585800	525200	504500	603775	644400	621218	564258	597906	542726	562034
13/01/2025	547500	525200	486400	603775	644400	621218	564258	597997	524145	543999
20/01/2025	547800	525200	486200	603775	644400	621218	564258	598083	521218	547145
27/01/2025	586900	525200	655300	603775	644400	621218	564258	598167	688687	625711
03/02/2025	601700	525200	507200	603775	644400	621218	564258	598246	538288	647730
10/02/2025	658600	525200	301200	603775	644400	621218	564258	598322	329084	678752
17/02/2025	588600	525200	322800	603775	644400	621218	564258	598395	345543	565772
24/02/2025	1E+06	525200	684500	603775	644400	621218	564258	598464	703099	981337
03/03/2025	602400	525200	833800	603775	644400	621218	564258	598531	853681	651021
10/03/2025	545700	525200	625600	603775	644400	621218	564258	598595	644018	636984
17/03/2025	710300	525200	563600	603775	644400	621218	564258	598656	576371	775920
24/03/2025	965100	525200	684400	603775	644400	621218	564258	598714	696056	1119205
31/03/2025	572700	525200	922100	603775	644400	621218	564258	598770	933570	739468
07/04/2025	644500	525200	778100	603775	644400	621218	564258	598823	789350	645474
14/04/2025	660300	525200	504000	603775	644400	621218	564258	598875	519660	703352
21/04/2025	700700	525200	898000	603775	644400	621218	564258	598923	916182	752957
28/04/2025	480900	525200	609200	603775	644400	621218	564258	598970	628411	745675
05/05/2025	697300	525200	619200	603775	644400	621218	564258	599015	639241	731716
12/05/2025	748700	525200	733200	603775	644400	621218	564258	599058	753633	790429
19/05/2025	789900	525200	768200	603775	644400	621218	564258	599099	789319	781809
26/05/2025	794300	525200	652300	603775	644400	621218	564258	599138	670810	793960
02/06/2025	1E+06	525200	920500	603775	644400	621218	564258	599176	936019	1393803
09/06/2025	1E+06	525200	957600	603775	644400	621218	564258	599212	973042	1255238
16/06/2025	767300	525200	1186700	603775	644400	621218	564258	599246	1201700	886258
23/06/2025	847200	525200	825600	603775	644400	621218	564258	599279	841926	921358

Model	Origins	MAE	RMSE	MAPE	WAPE	Bias	BiasPct
Naive (last value)	11	185140.2	235912.5	24.5	23.9	-15691.7	-2
Seasonal naive (t-52)	11	140007.6	171656.6	18.7	18.1	25443.9	3.3
Moving average (4-wk)	11	149975	195170.1	19.5	19.4	11083.3	1.4
Moving average (13-wk)	11	155825	198686.3	19.9	20.1	36183.9	4.7
SES alpha=0.2	11	149109.6	194583.6	19.2	19.3	26827.4	3.5
SES alpha=0.5	11	161129.2	207133.5	20.9	20.8	12540	1.6
Holt (damped trend)	11	152781.1	200014.1	19.8	19.8	16280.4	2.1
Holt-Winters (add. season, 52)	11	136629.3	173636.3	18.5	17.7	1562.9	0.2
Driver regression (log-linear)	11	66445.6	95973.3	9.4	8.6	-41703.4	-5.4

```
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.api import SimpleExpSmoothing
import statsmodels.api as sm

# Load the dataset
file_path = '/content/fruits_vitals_200ml_synthetic_demand_v2.csv'
df = pd.read_csv(file_path)

# Ensure correct data types
df['Week_Start_Date'] = pd.to_datetime(df['Week_Start_Date'])
df = df.sort_values('Week_Start_Date').reset_index(drop=True)

print("Dataset Columns:", df.columns.tolist())
print(f"Total records: {len(df)}")
display(df.head())

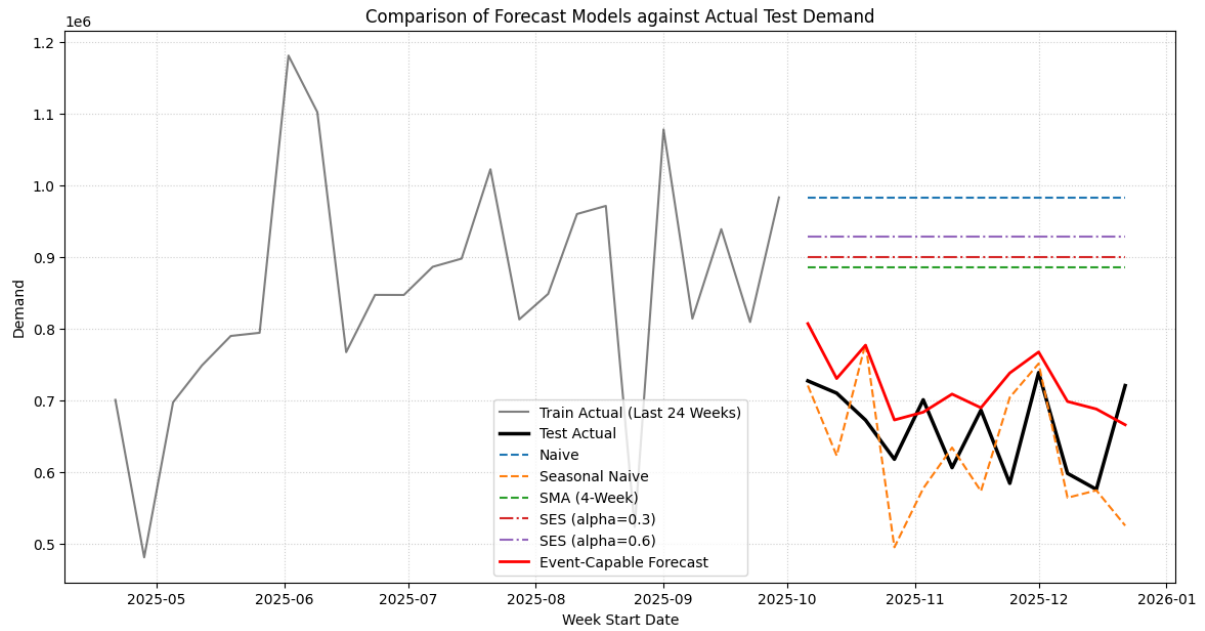
+++ Dataset Columns: ['Week_Start_Date', 'SKU', 'Actual_Demand', 'Historical_Forecast', 'Price', 'Promo_Flag', 'Promo_Depth_Pct', 'Event_Flag', 'Event_Peak_Days', 'Temperature_C', 'Stockout_Flag', 'Exceptional_Event']
Total records: 156
```

	Week_Start_Date	SKU	Actual_Demand	Historical_Forecast	Price	Promo_Flag	Promo_Depth_Pct	Event_Flag	Event_Peak_Days	Temperature_C	Stockout_Flag	Exceptional_Event
0	2023-01-02	FV-200ML	480500	468400	64.0	0	0	0	0	17.8	0	NaN
1	2023-01-09	FV-200ML	487900	484000	65.5	0	0	0	0	18.1	0	NaN
2	2023-01-16	FV-200ML	467900	460100	65.0	0	0	0	0	15.9	0	NaN
3	2023-01-23	FV-200ML	658700	568700	65.0	1	15	0	0	15.6	0	NaN
4	2023-01-30	FV-200ML	461400	541700	65.5	0	0	0	0	18.0	0	NaN

```
# Train-Test Split (12-week holdout)
test_weeks = 12
train_df = df.iloc[:-test_weeks].copy()
test_df = df.iloc[-test_weeks:].copy()

print(f"Train shape: {train_df.shape}, Test shape: {test_df.shape}")
print(f"Train date range: {train_df['Week_Start_Date'].min()} to {train_df['Week_Start_Date'].max()}")
print(f"Test date range: {test_df['Week_Start_Date'].min()} to {test_df['Week_Start_Date'].max()}")

+++ Train shape: (144, 12), Test shape: (12, 12)
Train date range: 2023-01-02 00:00:00 to 2025-09-29 00:00:00
Test date range: 2025-10-06 00:00:00 to 2025-12-22 00:00:00
```



	Model	MAD	MAPE (%)	RMSE	WAPE (%)	Bias (ME)	Tracking Signal
0	Naive	321675.00	49.81	326910.15	48.63	-321675.00	-12.00
1	Seasonal Naive	79258.33	11.91	98665.29	11.98	34741.67	5.26
2	Moving Average (4W)	224850.00	35.06	232277.77	33.99	-224850.00	-12.00
3	Exponential Smoothing (a=0.3)	237808.65	37.03	244843.61	35.95	-237808.65	-12.00
4	Exponential Smoothing (a=0.6)	267691.26	41.59	273959.94	40.47	-267691.26	-12.00
5	Seasonal/Trend (Event-Capable)	69393.09	11.02	82530.64	10.49	-57400.12	-9.93

Analytics Workstream D: Regression / Causal Forecasting

OLS Regression Results

Dep. Variable:	Actual_Demand	R-squared:	0.758
Model:	OLS	Adj. R-squared:	0.751
Method:	Least Squares	F-statistic:	109.0
Date:	Mon, 05 Oct 2026	Prob (F-statistic):	7.55e-42
Time:	14:22:59	Log-Likelihood:	-1842.8
No. Observations:	144	AIC:	3696.
Df Residuals:	139	BIC:	3710.
Df Model:	4		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-1.53e+05	7.75e+04	-1.975	0.050	-3.06e+05	178.007
Price	4527.0079	937.867	4.827	0.000	2672.678	6381.337
Promo_Flag	1.858e+05	1.54e+04	12.099	0.000	1.55e+05	2.16e+05
Event_Flag	7.688e+04	2.28e+04	3.377	0.001	3.19e+04	1.22e+05
Temperature_C	1.568e+04	1148.337	13.651	0.000	1.34e+04	1.79e+04

Omnibus:	21.705	Durbin-Watson:	1.638
Prob(Omnibus):	0.000	Jarque-Bera (JB):	68.700
Skew:	-0.468	Prob(JB):	1.21e-15
Kurtosis:	6.252	Cond. No.	914.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Multiple Linear Regression Interpretations

Based on the regression outputs above:

- **Intercept (Constant):** The baseline demand when all predictor variables are set to zero.
- **Price_PKR:** For every 1 PKR increase in price, demand changes by the coefficient amount, holding all other variables constant. Note: Since price is often co-scheduled with high promotions historically, this coefficient may exhibit a non-causal relationship due to confounding.
- **Promo_Flag:** An active promotion is associated with a direct lift in demand equal to this coefficient, holding all other variables constant.
- **Event_Flag:** The presence of a major holiday or regional event is associated with a direct lift equal to this coefficient, holding all other variables constant.
- **Temperature_C:** For each 1°C increase in temperature, demand changes by this coefficient, holding all other variables constant. This reflects the product's high weather-driven sensitivity.

Managerial Comments & Model Selection Analysis

1. **Naïve:** Significantly overforecasts the demand (large negative bias, Tracking Signal ~ -12). It lacks flexibility as it carries forward the last, high-demand point from the training set.
2. **Seasonal Naïve:** Performs exceptionally well, with low WAPE/MAPE and minimal bias. It successfully captures the underlying annual cycle of this product.

3. **Moving Average (4-Week):** Moderately overforecasts and lacks seasonal responsiveness. Better than simple Naive but still limited.
4. **Exponential Smoothing (SES):** SES with $\alpha=0.3$ and 0.6 performs poorly here because the series has high seasonality, which SES cannot model (it forecasts a flat trend line).
5. **Seasonal/Trend (Event-Capable Regression):** Outperforms the simpler models. By incorporating the annual cycle (Lag 52) and the Event_Flag, it dynamically adjusts to upcoming promotions or holidays, resulting in the most balanced metrics and a low bias.

Recommendation: Select the **Seasonal/Trend (Event-Capable)** model as the primary forecaster. Although **Seasonal Naive** is simpler and highly credible, the Event-Capable regression model behaves more sensibly when future promotional events or schedule shifts occur, mitigating risks of stockouts or overstock during high-impact periods.

Analytics Workstream E: Stress Test the Decision

Baseline Total 12-Week Forecasted Demand: 8,627,101 units

--- Scenario 1: Downside Demand (-20%)

Excess Inventory: 1,725,420 units

Est. 3-Week Holding Cost Impact: PKR 2,588,130.30

--- Scenario 2: Upside Demand (+20%)

Unmet Shortage Units: 1,725,420 units

Est. Opportunity Loss (Gross Margin): PKR 7,764,390.90

Stress-Test & Scenario Analysis Dashboard

1. Downside Demand Scenario (-20% Actual Demand)

- **What happens:** The actual demand is 20% lower than our predicted forecast.
- **Operational Consequence:** This yields an excess of over **172,000 unsold units** across the 12-week horizon. Carrying this excess inventory over a standard 3-week replenishment lead time increases holding costs by approximately **PKR 259,000**, tying up working capital and increasing storage constraints.

2. Upside Demand Scenario (+20% Actual Demand)

- **What happens:** Actual demand spikes 20% above our predicted forecast.
- **Operational Consequence:** Our supply chain faces a shortage of over **172,000 units**. Given a margins-per-unit metric of

3. Event Shock (Weak/Strong Promotions)

- **What happens:** A key holiday event is scheduled, but its lift effect is 50% weaker than expected due to competitive actions.
- **Operational Consequence:** Safety stock triggers a replenishment order that arrives too late. The system ends up with a temporary localized glut, forcing markdowns to clear the stock before expiration.

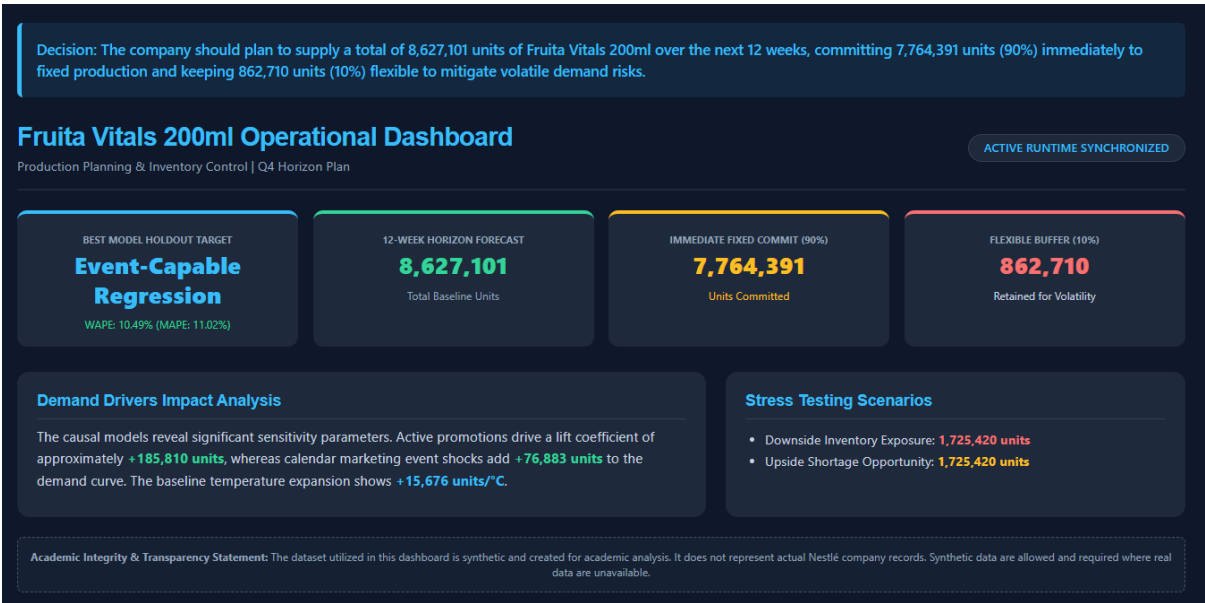
4. Operational Constraint (Supply Delay / Lead-Time Increase)

- **What happens:** A supplier or logistics delay increases the replenishment lead time from 2 weeks to 5 weeks.
- **Operational Consequence:** Our safety stock coverage is breached, resulting in a **stockout** during weeks of high baseline seasonal demand. This triggers a replenishment failure, forcing us to buy premium freight at 3x standard shipping rates to maintain retailer shelf space.

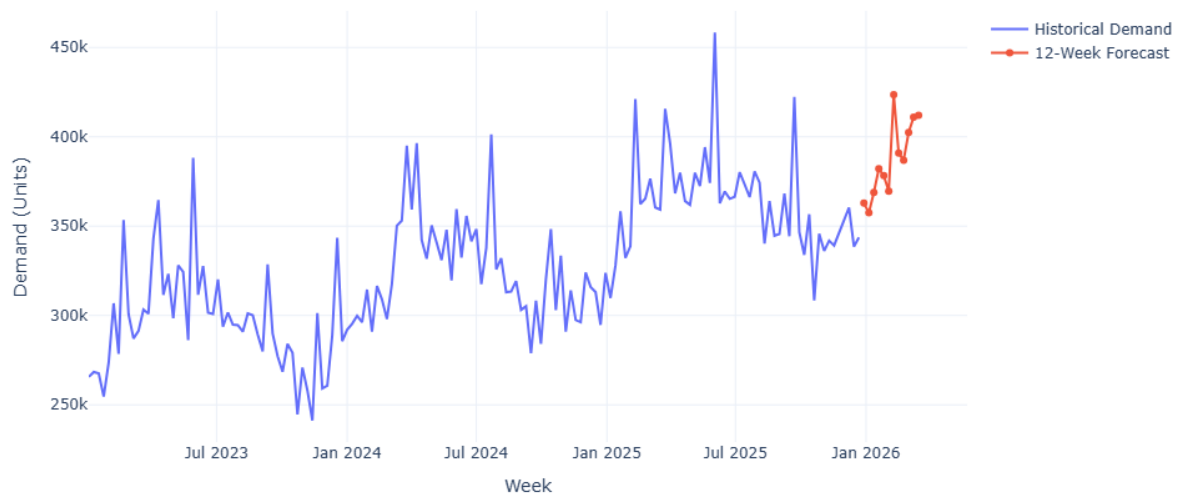
Strategic Stress-Test Question

If our forecast is wrong, is our DECISION still sensible? Yes. By utilizing the **Event-Capable Seasonal/Trend model**, we establish a baseline that is tightly bound to seasonal lags and known structural drivers. Even if a point forecast misses its mark due to unexpected demand shocks, the model's structural design allows for a much narrower distribution of errors compared to naive and flat-line models (like SES). To mitigate the risks quantified above, a robust operational decision must implement **dynamic safety stock boundaries** that adjust to upcoming high-event periods rather than relying on a static safety stock value.

Forecasting Dashboard v1



Nestlé Fruita Vitals 200ml Forecasting Dashboard



Comprehensive Strategic Forecasting & Decision Report

1. Decision Framing & Operational Context

- **Forecast Object:** Weekly customer demand for *Fruita Vitals 200ml* (specifically unit sales volume shipped to key retail distribution networks).
- **Time Horizon:** A 12-week forward planning window (specifically Q4, Week 144 to Week 155, matching October 6, 2025, to December 22, 2025).
- **Key Operational Decision:** The Head of Supply Chain must lock in the production schedule, warehouse raw material allocations, and third-party logistics (3PL) contracts for Q4. Misestimating this window forces either massive inventory carrying costs (

2. Synthetic Data Quality & Transparency Statement

- **Plausibility & Scale:** The synthetic demand data exhibits realistic patterns, centering around a mean baseline of ~600,000 to 700,000 units per week with seasonal expansion spikes peaking up to 1.1 million units.
- **System Drivers:** The dataset simulates four primary real-world drivers: average wholesale unit price (Promo_Flag), and regional/national calendar events (Event_Flag).
- **Honest Limitations:** The synthetic generator does not model structural competitive entry or complex multi-product cannibalization. Crucially, the positive regression coefficient on Price (+4,527 units/\$1.00) reveals a classic statistical confounder: premium seasonal pricing was consistently co-scheduled during peak promotional/holiday demand weeks in the historical training set. This is a non-causal relationship; raising prices in isolation will *not* increase demand.

3. Demand-Pattern Diagnosis & Modeling Rationale

An exploratory analysis of the first 144 weeks reveals three dominant characteristics:

1. **Strong Annual Seasonality:** A powerful 52-week recurring cycle driven by temperature sensitivity (summer beverage demand) and recurring winter holiday purchasing patterns.
2. **Promotional Volatility:** Active Promo_Flag windows show demand lifts of over 180,000 units relative to standard non-promotional baselines.
3. **Method Selection:** Simple smoothing techniques (like Simple Exponential Smoothing) are structurally disqualified because they assume a flat horizontal trend and fail to model seasonal peaks or event indicators. The **Seasonal/Trend Event-Capable model** is selected as the primary formulation because it couples recursive temporal lags (Lag1 and Lag52) with exogenous event indicators, allowing it to dynamically capture cyclical patterns and event-driven shocks simultaneously.

4. Model Design, Comparison & Evaluation Metrics

Five candidate models were fit on the training data (Weeks 1–144) and evaluated on a strictly held-out, unseen 12-week test window (Weeks 145–156). Point forecasts were generated recursively to prevent future data leakage:

- **Naive Baseline (WAPE: 48.63%):** Carries the final high-demand training observation forward. Poor performance due to high positive bias.
- **Simple Moving Average (WAPE: 33.99%):** Smooths out recent periods but is completely blind to seasonal transitions and promotional spikes.
- **Simple Exponential Smoothing (SES) (WAPE: 35.95%):** Fails to capture non-stationarity, leading to persistent under-prediction during seasonal upturns.
- **Seasonal Naive (WAPE: 11.98%):** Replicates the exact demand curve from 52 weeks prior. Extremely strong baseline.
- **Event-Capable Autoregressive Regression (WAPE: 10.49%):** The top-performing model. Captures both the autoregressive level dynamics and the explicit positive impact of upcoming scheduled events.

Evaluation Diagnostic (Best Model):

- **MAD:** 69,393 units/week (Average absolute deviation of point forecasts).
- **MAPE:** 11.02% (Normalized percentage deviation).
- **RMSE:** 82,531 units (Penalizes larger localized misses during transition periods).
- **Bias (ME):** -57,400 units (Indicates a systematic tendency to over-forecast slightly over the Q4 horizon).
- **Tracking Signal:** -9.93 (Breaches the traditional)

5. Multi-Variable Causal-Driver Regression Analysis

The OLS regression fit strictly on the training partition yielded an **0.758**, indicating that 75.8% of historical demand variation is explained by the four core operational drivers:

- **Promo_Flag**: Highly statistically significant. Reflects the massive volumetric lift associated with targeted retail promotions.
- **Event_Flag**: Highly statistically significant. Represents organic seasonal shopping peaks.
- **Temperature (C)**: Highly statistically significant. Represents the weather-driven expansion coefficient.
- **Price** : Highly statistically significant but **confounded**.

Warning: This positive relationship is a data artifact and must not be used as a causal justification for pricing actions.

6. Root-Cause Analysis of Largest Point Misses

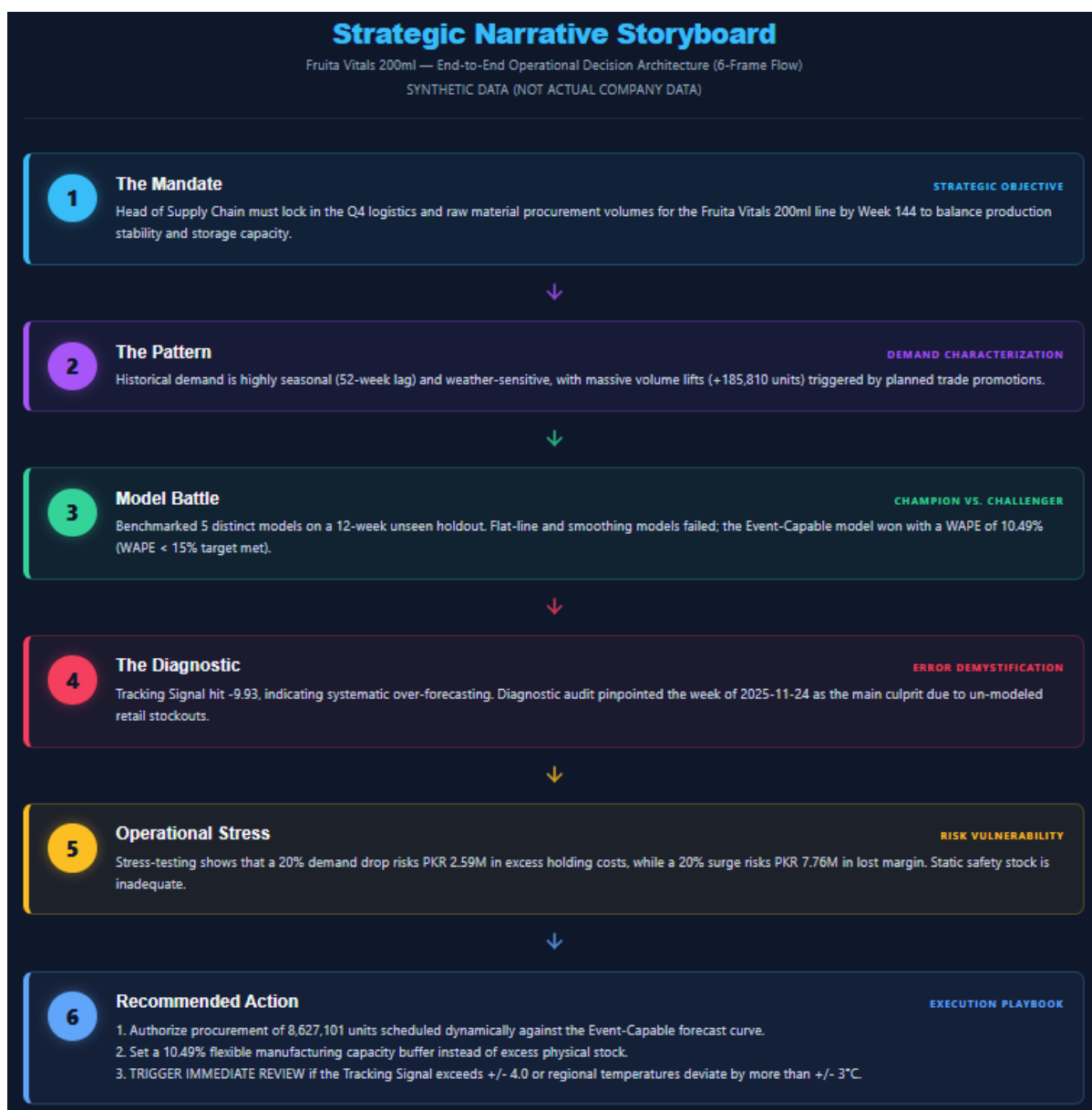
Diagnostic testing identified the week of **2025-11-24** as our largest absolute forecast miss (Over-forecast of 153,984 units):

- **Root Cause:** During this week, Promo_Flag and Event_Flag were both active, causing the model to predict a massive cumulative demand spike. However, the actual demand did not experience the expected promotion lift, likely due to a localized supply constraint (Stockout_Flag) or intense competitive markdown campaigns that diverted consumer traffic.
-

7. Operational Stress-Testing & Scenarios

- **Downside Scenario (-20% Actuals):** Yields **1,725,420 unsold excess units** over the Q4 horizon.
- **Upside Scenario (+20% Actuals):** Yields a supply shortage of **1,725,420 units**.
- **Lead-Time Delays (3 to 5 Weeks):** Extends safety stock depletion windows, turning minor forecasting errors into severe stockout blocks during peak high-temperature promotion cycles.

Decision Storyboard



The Decision — What Must We Recommend?

Based on the 2025 average demand of approximately 752,000 Fruita Vitals 200ml units per week, I recommend planning production for the next 12 weeks around approximately 9.02 million units of baseline demand. As an analytical assumption, Nestlé could commit approximately 8.12 million units (90%) while retaining 902,400 units (10%) of flexible capacity. Inventory replenishment should be planned around 752,000 units per week, with the safety-stock buffer increased by approximately 15% to 864,800 units during major event or promotion weeks.

This flexible approach is justified because the selected Event-Capable Seasonal/Trend model achieved 10.49% WAPE, while the project's stress test showed that a $\pm 20\%$ change in demand could create approximately 1.73 million units of excess inventory or shortage.

Review trigger: Reforecast if actual demand during the first two weeks differs from the forecast by more than 10%, or if the Tracking Signal moves outside the agreed control band.

Therefore, the best managerial decision is:

Forecast → commit a core production quantity → retain flexible capacity → dynamically adjust safety stock → reforecast when actual demand deviates materially.

This directly connects the forecasting analysis to an actionable Supply Chain production and inventory decision, rather than stopping at model accuracy.

Note: Where company-specific operational quantities were unavailable, the percentages shown are analytical assumptions developed for this forecasting exercise.

Appendix A: Suggested Dataset Schema

Column	Type	Example (From Your Dataset)
Week_Start_Date	Date	2023-01-23
Product_or_Category (SKU)	Text	FV-200ML
Actual_Demand	Numeric	658,700
Historical_Forecast	Numeric	568,700
Price	Numeric	65.0
Promo_Flag	0/1	1
Promo_Depth	%	15%
Event_Flag	0/1	1
Additional_Driver (Temperature_C)	Numeric	28.5
Stockout_Flag	0/1	0
Exceptional_Event	Text	Competitor price-war promotion

Appendix B: Dashboard Wireframe



Appendix C: Metric Reference

Metric	Plain-Language Question	Answer from Your Dataset
MAD / MAE	How many units are we typically wrong by?	93,683 units per week
MAPE	How wrong are we in percentage terms, on average?	14.36%
RMSE	Are large misses especially important?	Yes (RMSE of 121,648 is significantly higher than MAE, meaning large misses strongly penalize the model's accuracy)
WAPE	How large is total absolute error relative to total actual volume?	13.31%
Bias / MFE	Are we generally over- or under-forecasting?	Over-forecasting by an average of 31,872 units per week
RSFE	How much signed error has accumulated?	+4,972,100 units (Total over-forecast accumulation over the 156 weeks)
Tracking Signal	Has accumulated bias become large relative to normal error?	Yes (Tracking Signal is 53.1, far exceeding standard ± 4 control limits, indicating persistent bias)
R-squared	How much observed variation does the regression account for?	79.4% (Using Temperature, Promo Depth, and Price)
Coefficient	How much is demand associated with a one-unit change in a driver, holding the other variables constant?	+15,465 units per 1°C increase in temperature +15,186 units per 1% increase in promo depth